

Can Legal Citation-Verification Agents Be Talked Out of a Correct Answer?

Research Plan — NLLP 2026 Submission

Team course: UE23AM343BA2 — LLMs and Their Applications **Plan date:** July 13, 2026 |

Target submission: August 11, 2026 (self-imposed)

⚠ **Deadline check, please confirm with your team:** the official NLLP 2026 call for papers lists **27 August 2026** as the submission deadline (co-located with EMNLP 2026, OpenReview site ([EMNLP/2026/Workshop/NLLP](https://openreview.net/forum?id=EMNLP/2026/Workshop/NLLP))), not August 11. I've built this plan to your requested Aug 11 target since that gives you a real buffer before the actual deadline — but if Aug 11 was a mistaken date, you have 16 more days than this plan assumes, which you could use to strengthen the dataset or add a third strategy condition. Worth a 2-minute team check before you lock the calendar below.

1. Problem Statement

Legal AI hallucination has moved from an edge case to a documented, sanctionable, courtroom-scale problem. As of April 2026, researchers had documented over 1,300 court proceedings involving AI-fabricated content — invented cases, misattributed quotes, and citations that exist but don't support the claim attached to them — with roughly 500 involving licensed attorneys and escalating financial sanctions. Courts have responded with concrete process changes: several U.S. district courts now require disclosure of AI use in filings, many jurisdictions require signed certificates attesting that AI-generated citations were manually verified, and New York's court system adopted formal AI-use rules effective mid-2026.

The industry's answer to this has been to add a **verification layer**: an agent (or agentic pipeline) that checks citations before they reach a human or a filing. Recent work shows this helps — an agentic citation-verification system reliably catches non-existent cases, and a calibrated multi-agent debate pipeline with a "Skeptic" agent cuts fabricated-citation detections by 45%. But none of this work asks the question that matters once verification becomes a standard pipeline component: **is the verifier itself robust to being argued out of a correct judgment by another AI agent?** If a verification agent can be socially engineered — not just fooled by a good fake, but talked out of a correct rejection through confident, authoritative-sounding pressure — then the verification layer offers a false sense of security exactly where the stakes are highest.

Your existing experiment already produced the raw ingredients to test this: an "authority" persuasion strategy that mirrors real fabricated-authority claims, and a "nonsense" strategy (technically-worded but false argument using real terminology) that mirrors the single most dangerous documented failure mode in legal AI — a real, existing citation attached to the wrong content, which survives a cursory human review. The pivot is to stop treating

these as abstract behavioral curiosities and start treating them as a stress test of a specific, currently-being-deployed system pattern.

2. Research Questions

Primary RQ: When a legal citation/claim-verification agent has correctly identified a citation as valid or invalid, can an adversarial peer agent using social-influence strategies (authority claims, plausible-but-false technical arguments) cause it to reverse that correct judgment?

Secondary RQs:

- **RQ2 (size asymmetry):** Does susceptibility scale with the size gap between verifier and adversary — i.e., is a smaller/cheaper verifier model measurably less safe to deploy behind a larger adversarial (or compromised) peer agent?
 - **RQ3 (mitigation):** Does giving the verifier grounded retrieval access (the real case text/holding, not just parametric knowledge) at the moment of challenge reduce its susceptibility to authority-based pressure?
 - **RQ4 (failure type):** Is the verifier more vulnerable on "does this citation exist" checks or on the harder "does this citation actually support this claim" checks — the distinction that recent legal-AI red-teaming work has flagged as the real danger zone?
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3. What's Actually New Here

1. First test (that I could find in a same-day literature check) of **adversarial persuasion robustness specifically for legal citation-verification agents**, as opposed to general Q&A or citation-existence detection.
 2. Direct empirical link between a **named social-influence strategy taxonomy** (grounded in Cialdini-style persuasion principles, which you already have coded up) and a **real, currently regulated deployment pattern** (mandatory AI-citation-verification workflows).
 3. A **model-size-asymmetry safety result reframed as a deployment recommendation**: quantified evidence for whether it's safe to put a small/cheap model in the verifier role behind a larger generator or adversarial peer.
 4. An actual **mitigation test** (retrieval-grounding vs. ungrounded), which moves the paper from "here's a problem" to "here's a problem and here's how much a known fix actually helps" — this is the single biggest lever for standing out, since most adjacent work stops at documenting the vulnerability.
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4. Related Work Landscape (your working list — expand during Phase 0)

- **Legal hallucination & verification:** citation-verification benchmarks with injected hallucinations; calibrated multi-agent debate ("Skeptic" challenge) pipelines for legal AI trustworthiness — closest neighbors, but focused on *detection/auditing*, not *adversarial robustness of the detector itself*.
- **Multi-agent persuasion/conformity (general domain):** confidence-weighted persuasion override metrics in LLM debate; conformity-driven prompt-injection attacks on debate systems; work decomposing "spontaneous instability" (models flip on mere reconsideration) from genuine conformity — **this is the paper that motivates your control condition below, cite it directly.**
- **Sycophancy / answer stability:** large-scale counterargument-flip benchmarks across frontier models and many subjects; domain-specific (medical) sycophancy risk work published in a clinical journal — evidence this general question is being taken seriously by strong venues, but not in legal citation verification specifically.
- **Real-world grounding:** legal-tech and legal-press reporting on sanctions, court rules, and disclosure requirements — use for your introduction's motivation, not as technical related work.

Action item, Phase 0: do a proper search pass on Semantic Scholar / ACL Anthology / arXiv for "legal citation verification adversarial," "multi-agent debate legal hallucination," and "sycophancy legal AI" — confirm nothing published in the last 4–6 weeks scoops this specific framing.

5. Experimental Design

5.1 Task reframing

- **Target → renamed "Verifier."** Given a claim + citation, it must judge: valid / invalid / unsupported.
- **Manipulator → renamed "Adversary."** A peer agent in an agentic drafting pipeline pushing the Verifier to reverse a correct judgment.
- Conversation structure stays the same shape as your original notebook (multi-turn, confidence-scored), just re-labeled and re-grounded.

5.2 Dataset (biggest bottleneck — start Day 1)

Target: **90–120 items** (realistic given your timeline; 150 is a stretch goal, not a requirement).

- **Real, valid citation items:** pull from a public case law source (CourtListener, Harvard's Caselaw Access Project) — objectively checkable, no legal expert required for ground truth.
- **Fabricated citation items:** invented case names/citations that don't exist — checkable programmatically.

- **Real-citation-wrong-content** items (your highest-value category): a real, existing citation paired with a claim it does *not* support — this is the hardest and most realistic hallucination type, and it needs a human pass to construct well.
- **Quality control:** you don't need a lawyer to validate all 120 items, but get one qualified reviewer (law student contact, or a team member with legal coursework) to spot-check ~25-30 items for a validity sanity check — note this explicitly as a limitation if you can't get more.

5.3 Conditions (narrowed from 5 strategies to 2, +1 optional)

- **Authority** (keep as-is — directly maps to fabricated-authority claims)
- **Fabricated-plausible-citation** (repurposed "nonsense" strategy — now framed as the real-citation-wrong-content attack, not generic technobabble)
- *Optional if time allows:* **Consensus** (maps to multiple agents in a pipeline converging on a shared error) — cut first if behind schedule.
- **Control condition (new, required):** no-challenge reconsideration turn — Verifier is simply asked to re-confirm its answer with no adversarial content, to separate genuine persuasion from spontaneous instability.

5.4 Models — ⚠️ verify before you build anything

Your original notebook used `llama-3.3-70b-versatile` and `llama-3.1-8b-instant` on Groq. I found **conflicting signals** on whether Groq has deprecated these exact model IDs — some current docs say they're deprecated in favor of `openai/gpt-oss-120b` / `openai/gpt-oss-20b`, other current pricing pages still list them as active. **Day 1 task: hit**

<https://api.groq.com/openai/v1/models> yourself and get the ground truth before you build Phase 2 around model IDs that might 404 mid-run (this happened to you once already with the rate-limit issues — don't let a deprecated model be the second version of that problem).

- **Family 1 (keep):** Llama pair if still active, or `gpt-oss-120b` / `gpt-oss-20b` if not.
- **Family 2 (add for RQ2 generalization):** Qwen (Groq now hosts a Qwen3 series) as your second model family — gives you a large/small pair in a different lineage.

5.5 Metrics

- Flip rate (correct → incorrect judgment), by condition and model pair
- Confidence delta (self-reported, as before)
- **Calibrated stability check (new):** resample each post-challenge answer 3-5x at temperature > 0 and report agreement rate as a second, less gameable stability measure
- Mitigation effect size: flip rate with grounded retrieval context vs. without, same items, paired comparison

5.6 Mitigation experiment (RQ3) — keep this simple given time constraints

Don't build a live retrieval/search pipeline — you don't have time. Instead: for each dataset item, pre-fetch the real case holding/syllabus text once during dataset construction, and inject it as static "retrieved context" in the grounded condition vs. omit it in the ungrounded condition. This tests the mechanism (does grounding help) without requiring a working search tool.

6. Methodological Fixes Checklist (carried over from earlier discussion)

- ☐ No-challenge control condition included
- ☐ Fixed turn budget across all conditions (pick 2, document why)
- ☐ Self-reported confidence + resampling-based stability measure
- ☐ Second model family added
- ☐ Randomized/varied decoy placement (not always the same slot)
- ☐ Grounded-vs-ungrounded mitigation comparison run on identical item set (paired design)

7. Timeline — July 13 → August 11 (27 research days + 3 writing days)

Phase	Dates	Days	Deliverable
0. Scope lock	Jul 13–15	3	Final RQ wording, task reframing doc, dataset schema, confirm Groq model availability
1. Dataset build	Jul 16–22	7	90–120 verified items (valid / fabricated / real-citation-wrong-content), spot-checked by 1 reviewer
2. Harness build + pilot	Jul 23–27	5	Updated codebase: control condition, resampling-based confidence, 2 model families, static retrieval-grounding injection; pilot on ~15 items to catch bugs
3. Main experiment runs	Jul 28–Aug 2	6	Full run: 2–3 strategies × control × 2 model pairs × 2 model families
4. Mitigation experiment	Aug 3–5	3	Grounded vs. ungrounded paired run on same items
5. Analysis	Aug 6–8	3	Stats tests, figures, results tables, decide final narrative

Phase	Dates	Days	Deliverable
6. Writing	Aug 9-11	3	Full draft → internal review → format/anonymize/submit

Day 1 (today, Jul 13) concretely: confirm Groq model IDs, assign the dataset-building task, draft the reframed problem statement/RQ into your actual paper doc (this document → seed it), and do the 30-minute "has anyone scooped this" search pass.

8. Suggested Team Split (adapt to your actual team size — your notebook shows a 3-person split by strategy last time)

- **Person A — Dataset:** sourcing, fabrication, wrong-content pairing, coordinating the legal spot-check
- **Person B — Infra/harness:** control condition, resampling confidence, second model family, retrieval-grounding injection
- **Person C — Analysis/writing lead:** stats plan now (so Phase 5 isn't a scramble), related-work deep dive, drafting skeleton in parallel with Phases 1-4

9. If You Fall Behind: Cut in This Order

1. Drop the optional **Consensus** condition — keep Authority + fabricated-citation only.
2. Reduce dataset from 120 → 80 items.
3. Drop the second model family — report size asymmetry within one family only, note cross-family generalization as future work.
4. Reduce resampling from 5x to 3x, or apply it only to flipped cases rather than all cases.
5. **Cut last, if at all:** the mitigation experiment (Section 5.6). It's your strongest differentiator — if time is short, shrink it (fewer items) rather than cutting it entirely.

10. Submission Logistics (NLLP 2026)

- **Format:** official ACL style templates (LaTeX or Word), *ACL-style formatting guidelines
- **Length:** 8 pages for long paper (4 for short), plus one extra page after the limit reserved for an optional ethics/broader-impact statement — given your topic (legal harm, safety), include this section
- **Review:** double-blind — anonymize the submission, and if you post to arXiv first, don't self-cite it in the anonymized version
- **Submission site:** OpenReview, [EMNLP/2026/Workshop/NLLP](https://openreview.net/forum?id=EMNLP/2026/Workshop/NLLP)

- **Archival option:** you can submit as archival (published in ACL Anthology) or non-archival (abstract only) — archival is almost certainly what you want for a CV-building first paper
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11. Immediate Next Steps (today)

1. Verify Groq's current model catalog — 10 minutes, unblocks everything else.
2. Confirm the Aug 11 vs Aug 27 deadline question with your team.
3. Assign Phase 1 (dataset) owner and get case-law source access sorted today — this is the long pole.
4. Do the scooping-check literature search (30 min) before investing further.